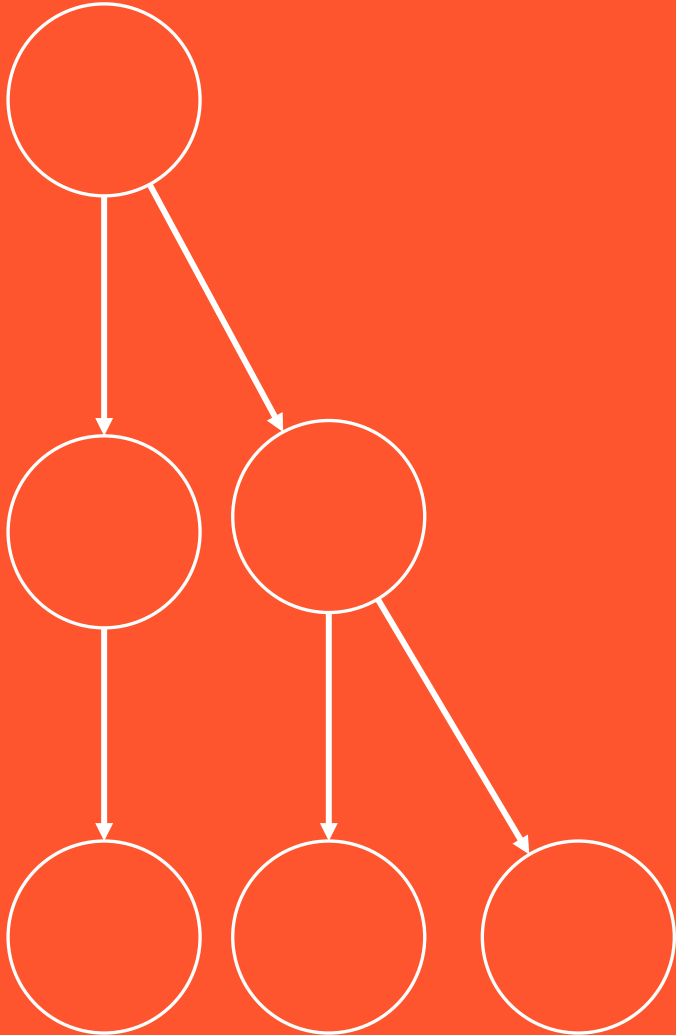


# BCOG 100

Learning and Plasticity

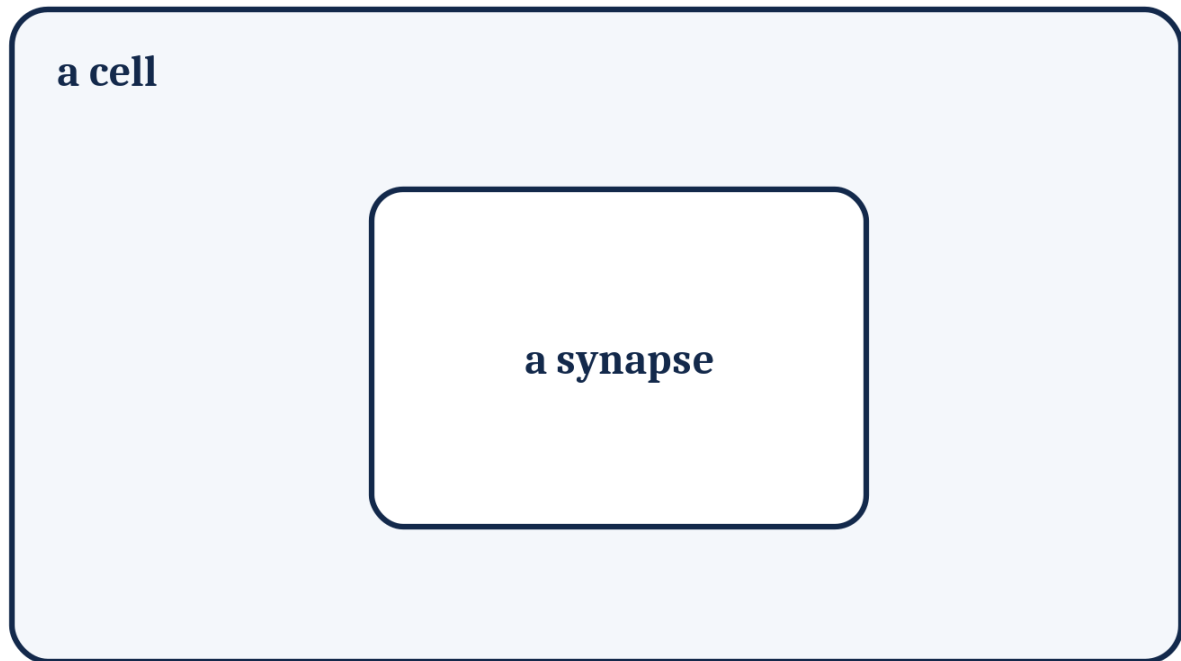
Lecture 5.3 — What Physically Changes, and What Gates It



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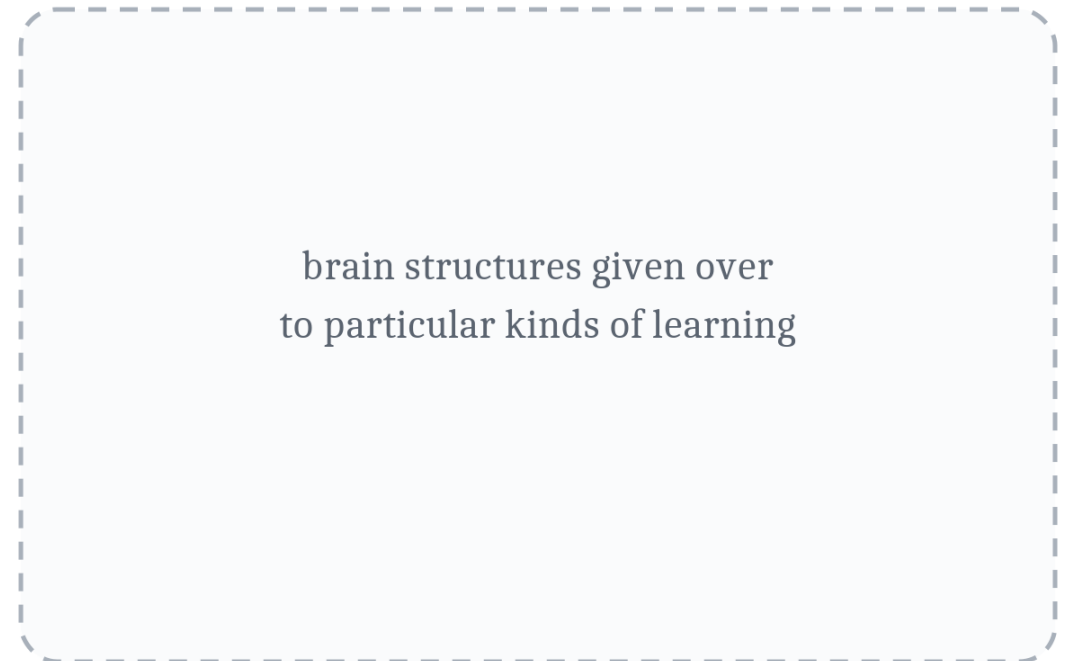
# One Principle, and Where This Account Stops

Underneath everything in this lecture: **a lasting change that is a function of activity**. What varies is where the change is made and what regulates it. The account stops at the synapse and the cell.



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**THIS LECTURE**



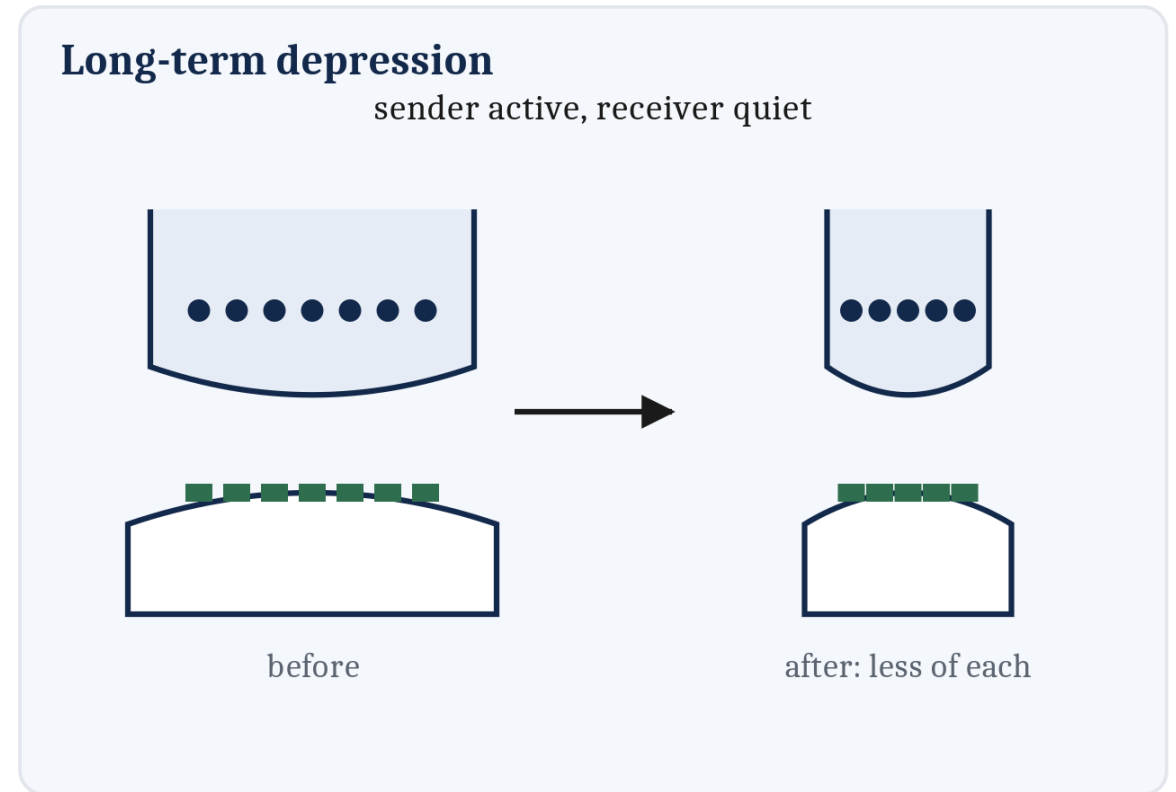
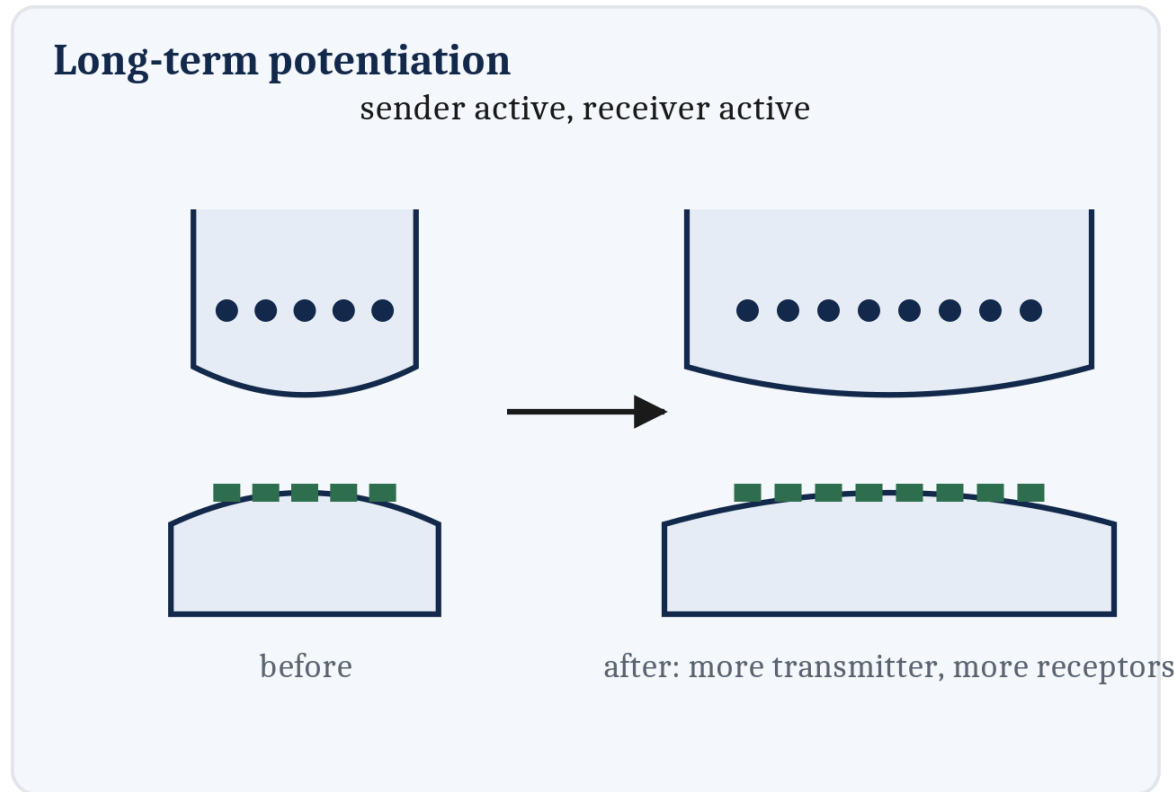
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**MODULE 6**

*Before there were brain systems for learning, there was learning at single synapses.*

# Strengthening a Synapse, and Weakening One

Fire together repeatedly and the synapse strengthens, and stays stronger: **long-term potentiation**. Sender active while the receiver stays quiet, and it weakens: **long-term depression**.

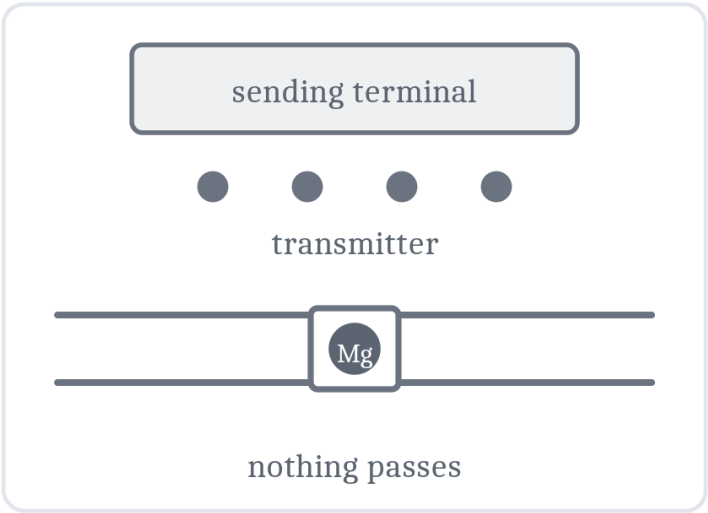


The pair together is what lets a weight move both ways.  
*A synapse that could only strengthen would be a one-way ratchet.*

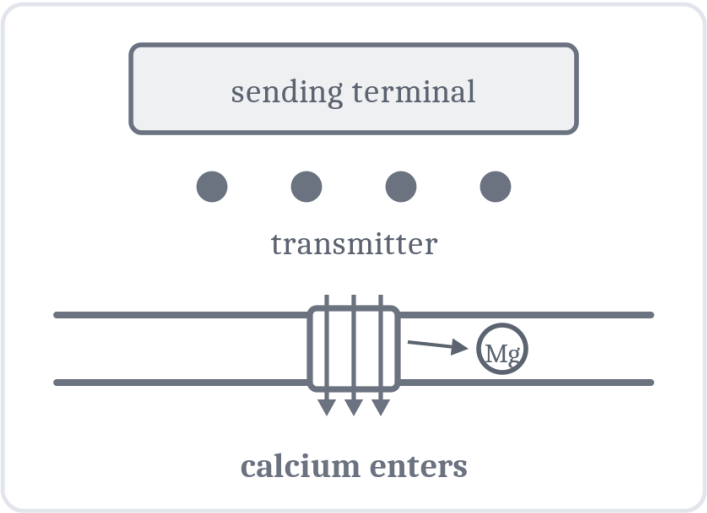
# The NMDA Receptor Is an AND Gate

The **NMDA receptor** opens only when transmitter is bound *and* the receiving cell is already depolarized. That is **coincidence detection**, and here the name is the mechanism.

A truth table, built out of chemistry



cell not depolarized: blocked



cell depolarized: open

| transmitter | depolarized | calcium    |
|-------------|-------------|------------|
| no          | no          | no         |
| yes         | no          | no         |
| no          | yes         | no         |
| yes         | yes         | <b>yes</b> |

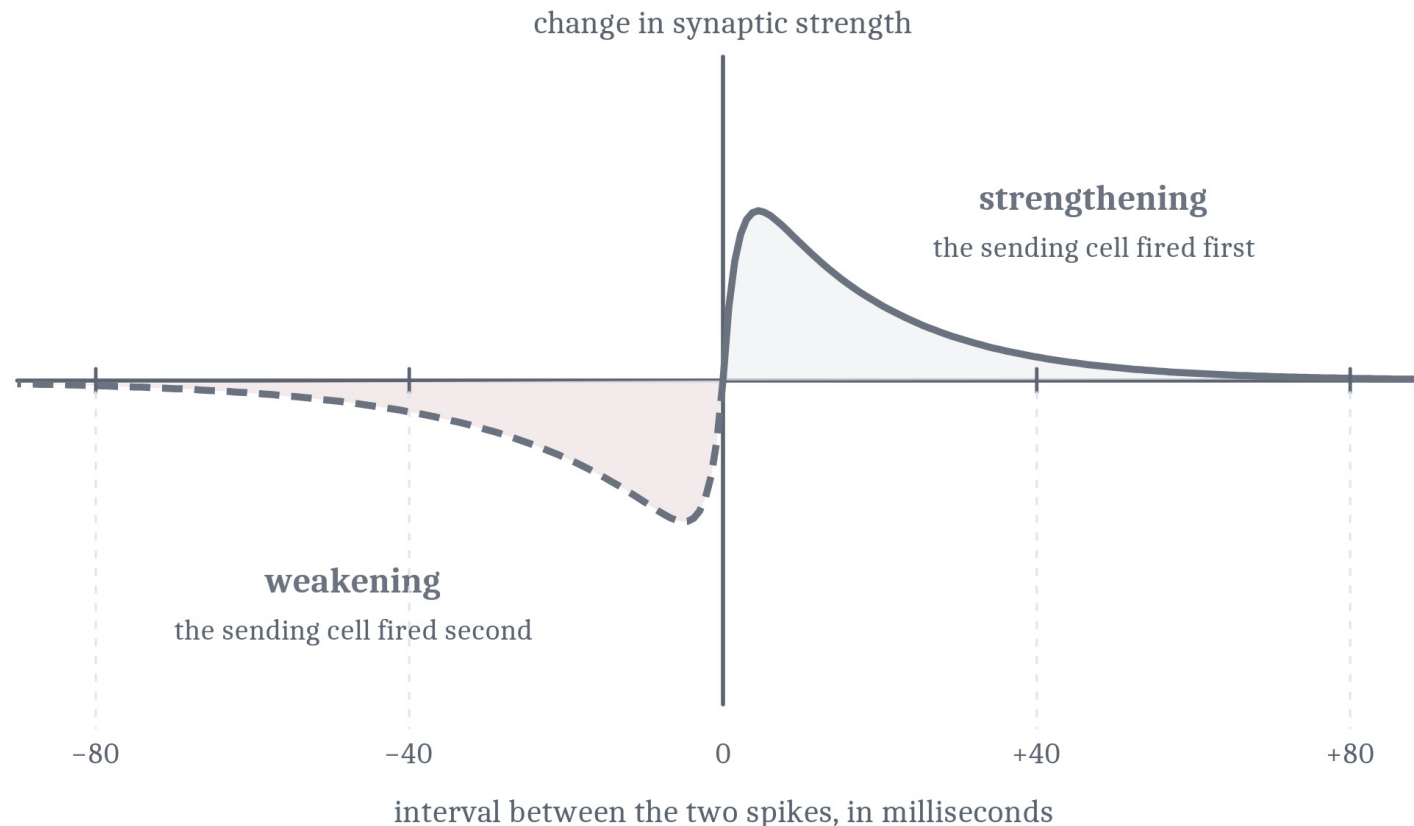
AND

either condition alone does nothing; only both together open the channel

# Timing, and Which One Came First

Sender fires just before the receiver: the synapse strengthens. Just after: it weakens. The window is a few tens of milliseconds — **spike-timing-dependent plasticity**.

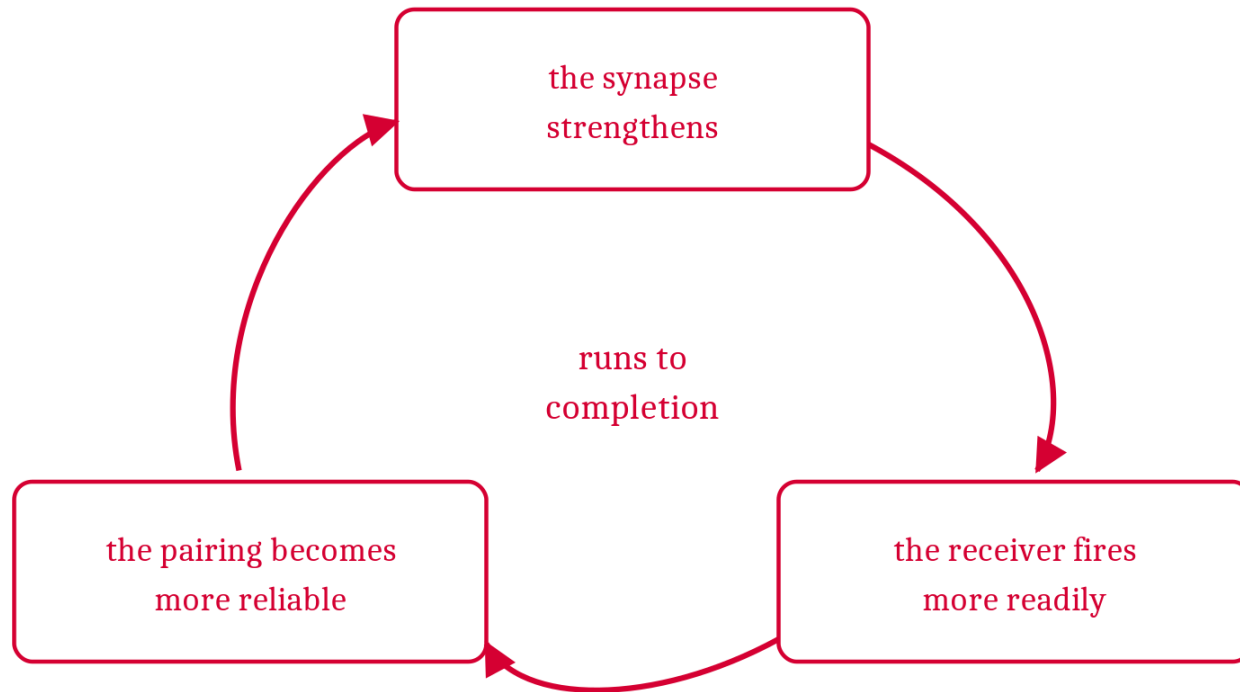
Whether a synapse strengthens depends on which cell fired first



outside a window a few tens of milliseconds wide, very little happens in either direction

# A Coincidence Rule Runs Away, and Biology Bounds It

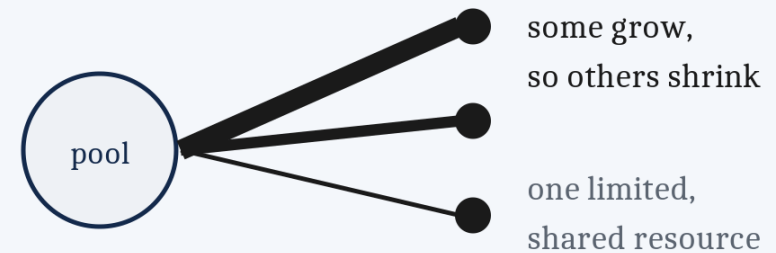
Strengthening what is already strong is positive feedback. Two things bound it: long-term depression, and **synaptic competition** for a limited, shared resource. Hebbian learning *can* weaken a connection.



## Brake 1: long-term depression

weakens a synapse directly

## Brake 2: synaptic competition

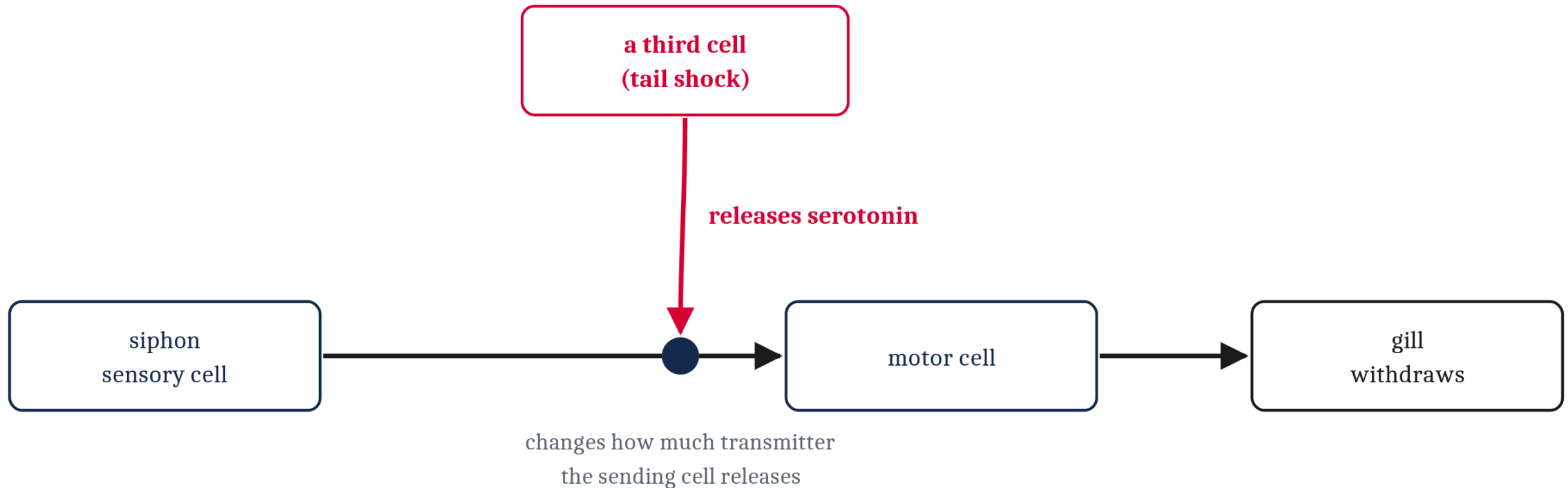


Hebbian learning can weaken a connection, two different ways.

*An input that stops carrying signal loses ground it already held.*

# *Aplysia*, and a Third Cell

One reflex in a sea slug carries habituation, sensitization and classical conditioning, each traced to identified synapses. Sensitization comes from a **third cell** releasing serotonin onto the pair.



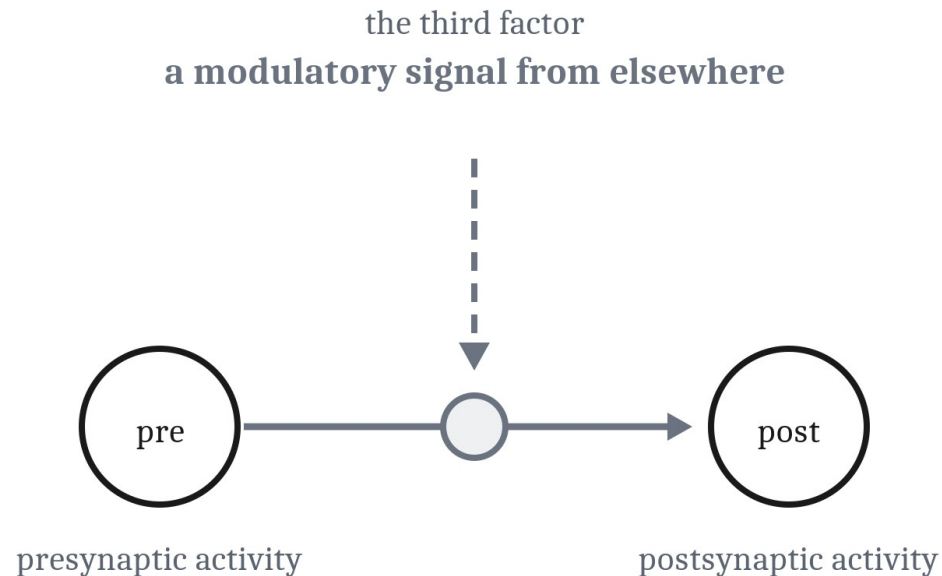
Neither cell in the pathway changed on its own. A signal from outside the pair did it.

*Aplysia*: about twenty thousand neurons, many large enough to see.

# The Three-Factor Rule

Presynaptic activity and postsynaptic activity decide which synapses are eligible. A third signal, broadcast from elsewhere, decides whether they change — and in **which direction**.

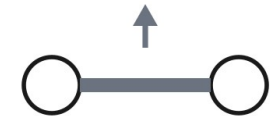
**Two cells make a synapse eligible. A third signal decides what happens.**



**together, these make  
the synapse eligible**

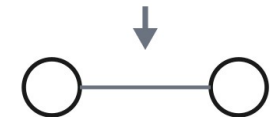
the third factor can reverse the direction of the change, not merely scale its size

a signal that says  
better than expected



**strengthened**

a different  
modulatory signal



**weakened**

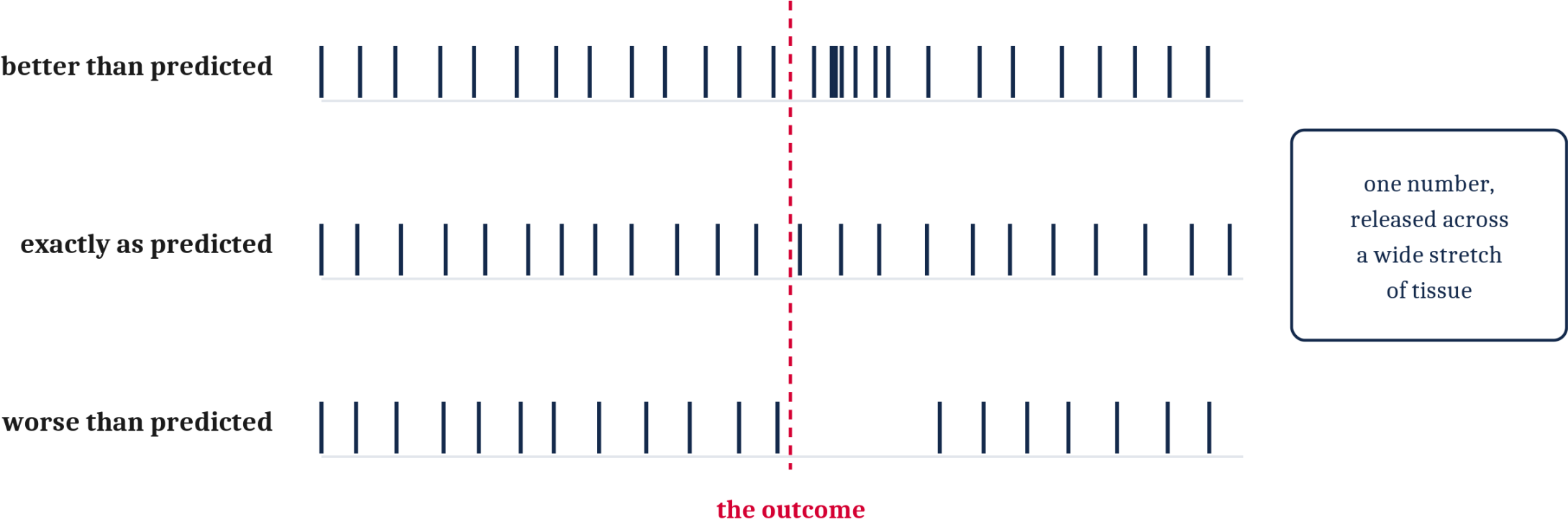
no signal arrives  
inside the window



**unchanged**

# Dopamine Reports Better and Worse Than Expected

Above baseline when an outcome beats the prediction, below it when the outcome falls short, unchanged when it arrives as predicted: the **reward prediction error**, broadcast.



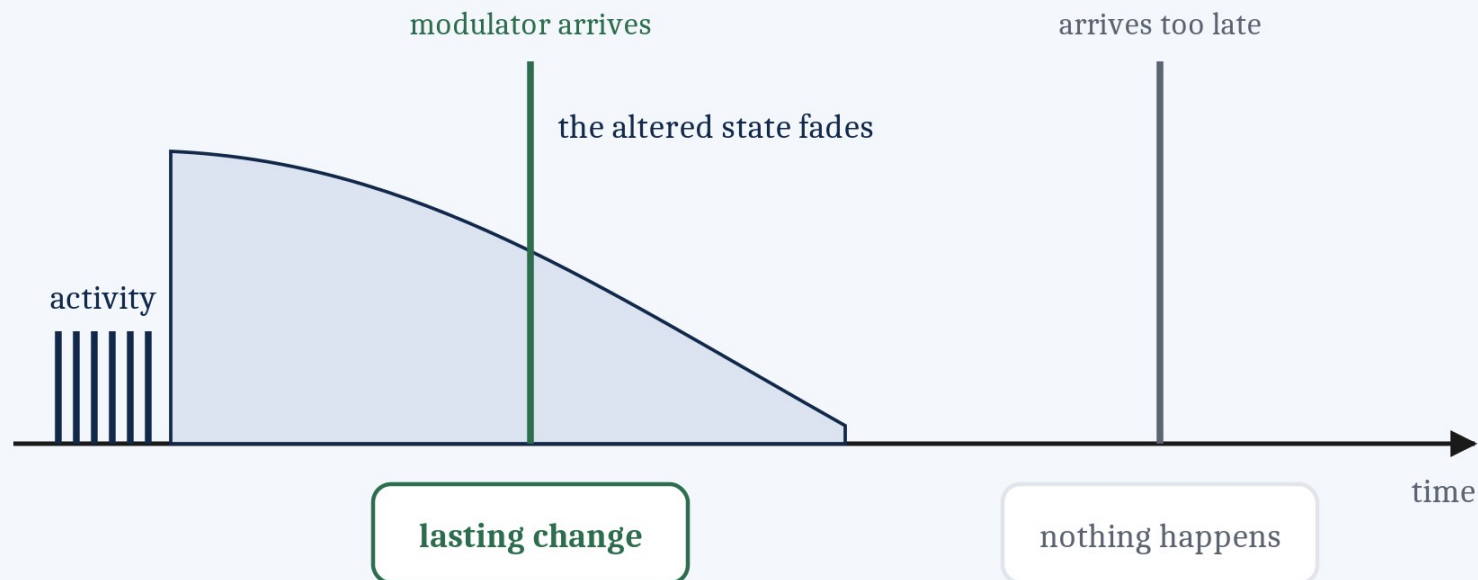
*Above baseline, at baseline, below baseline: the reward prediction error.*

Schematic. The recordings were made in vertebrates, largely primates.

# The Trace Made Physical, and the Other Signal From Elsewhere

A synapse that was just active is left in an altered state that fades over seconds. A modulatory signal inside that window makes the change last; after it, nothing happens.

## At a synapse, over a few seconds



## The other signal

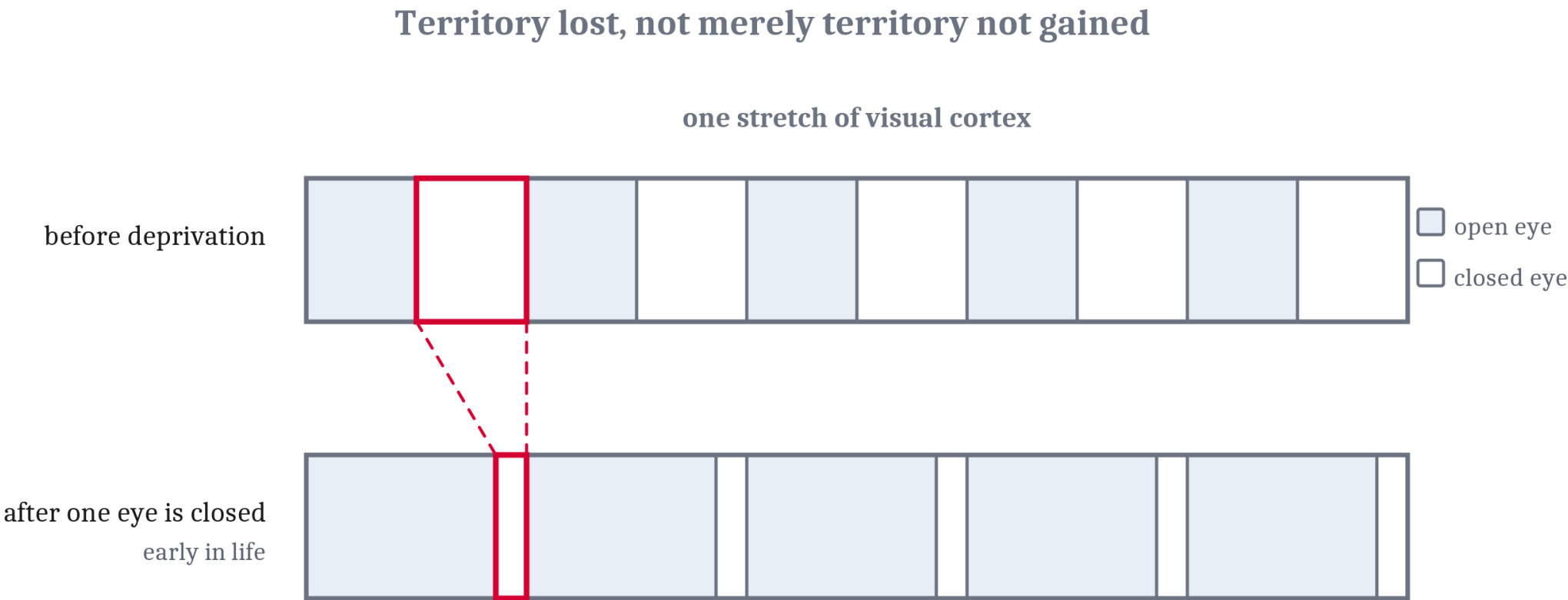


The thing the algorithm needed is the sort of thing a synapse can be.

*That does not show the algorithm is correct.*

# Learning Has a Schedule

Connections are made in profusion (**synaptogenesis**), then cut back by activity (**pruning**). A **critical period** is a window when experience has an outsized, lasting effect on wiring.



the same stripe, before and after

the closed eye holds less than it already held, and does not recover the difference

# Critical Periods: What the Evidence Supports, and What It Does Not

A window that closes changes what later learning has to work with. It does not end learning.

## Supported

Early experience has effects on some capacities that are large, specific, and hard to reverse

The deprived eye and infant phoneme loss are the clearest cases

Recovery from early deprivation is usually partial

Adults learn second languages

## Not supported

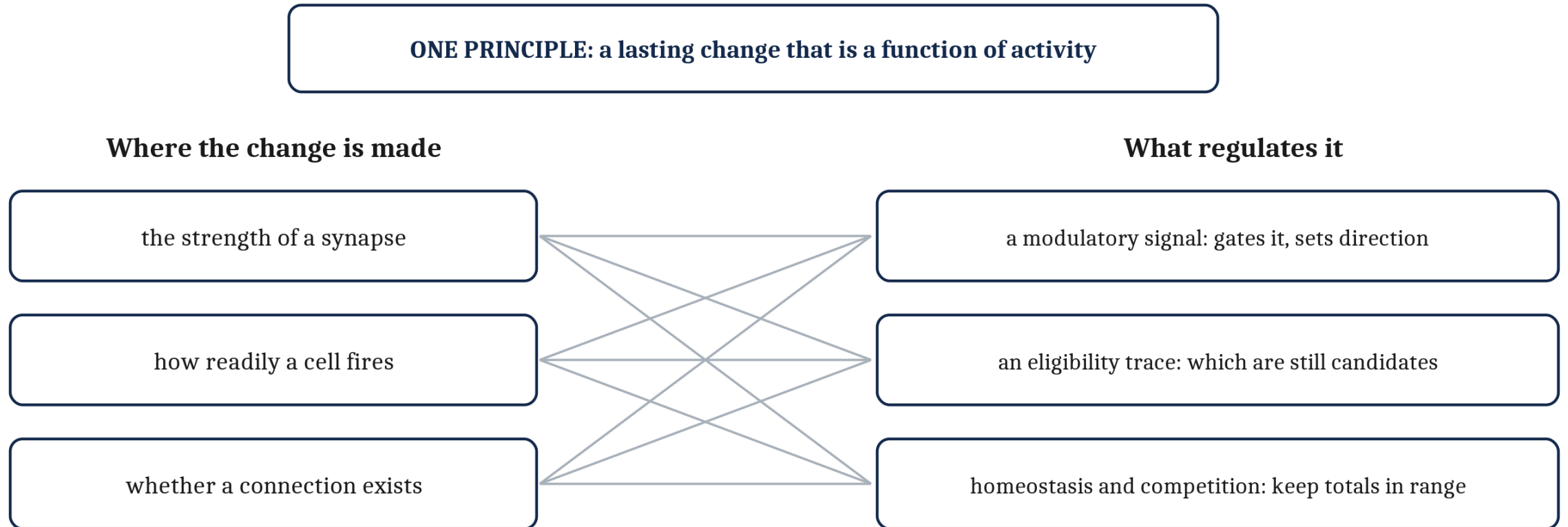
Development is a series of windows that slam shut

Missing a window means a capacity is gone for good

Extra stimulation during a window produces a better capacity later

# One Principle, Several Loci, Several Regulators

Three places to change: synaptic strength, **intrinsic excitability**, whether a connection exists. Three regulators: a modulatory signal, an eligibility trace, **homeostatic plasticity** with competition.



*Every mechanism in this lecture is one of each.*

# A Synapse Changed and Stayed Changed. Is That a Memory?

A **disposition** is committed when it is laid down. A **record** is kept without its use being settled, and what it is for is decided at retrieval. Learning produces dispositions.

## A disposition

## A record

Committed at encoding

Yes — what the change will do is fixed by the change

No — what it is for is settled later

A separate retrieval step

Not needed; the change acts whenever the situation recurs

Required, and it is where the use gets decided

What it can report

What tends to happen

What happened, on a particular occasion

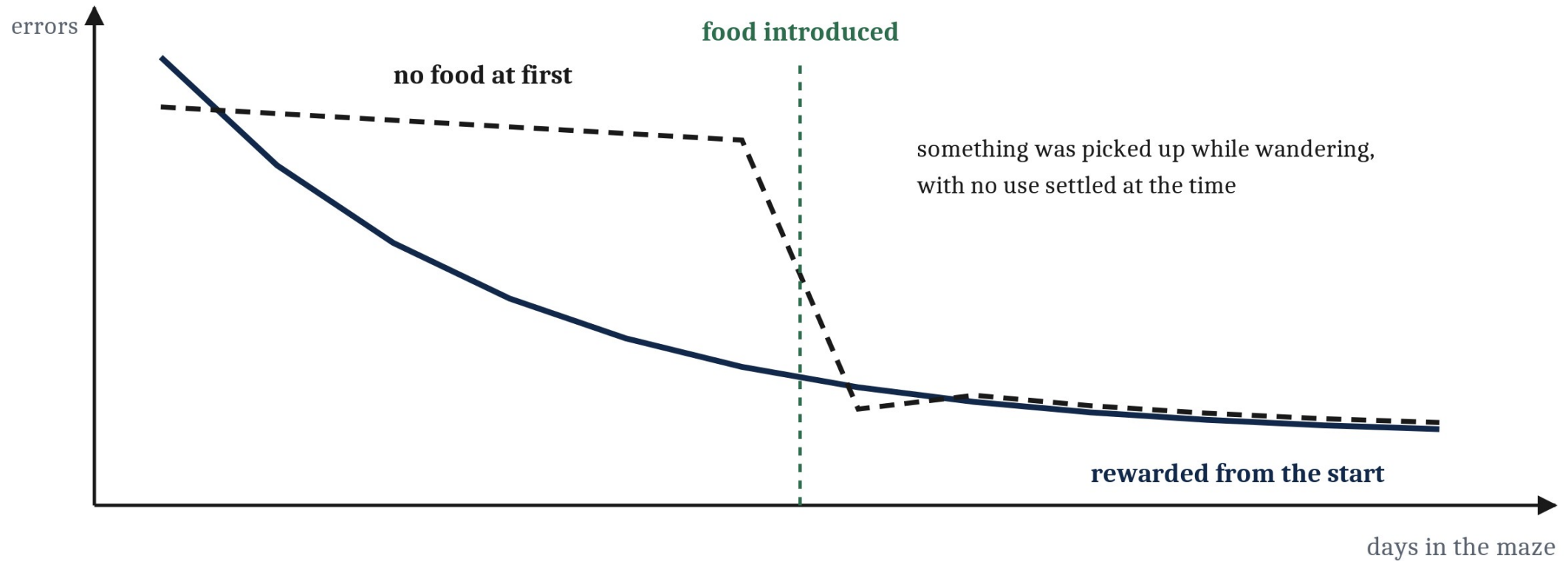
Where it is taken up

Module 5

Module 10

# Where a Disposition Stops Being Enough

In sensory preconditioning and latent learning, something is kept before anything has settled what it is for. That is where records begin, and Module 10 takes them up.

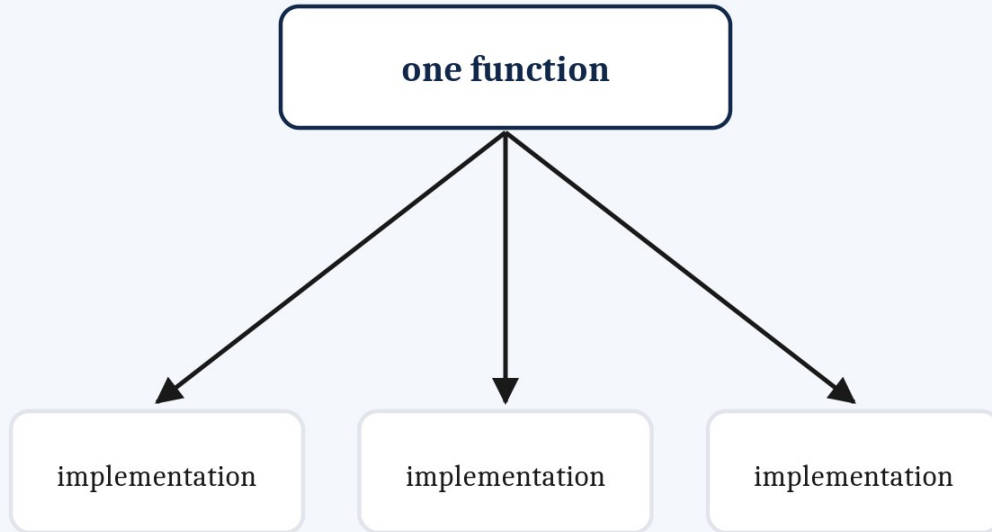


*Schematic, not data: the shape of the latent-learning result.*

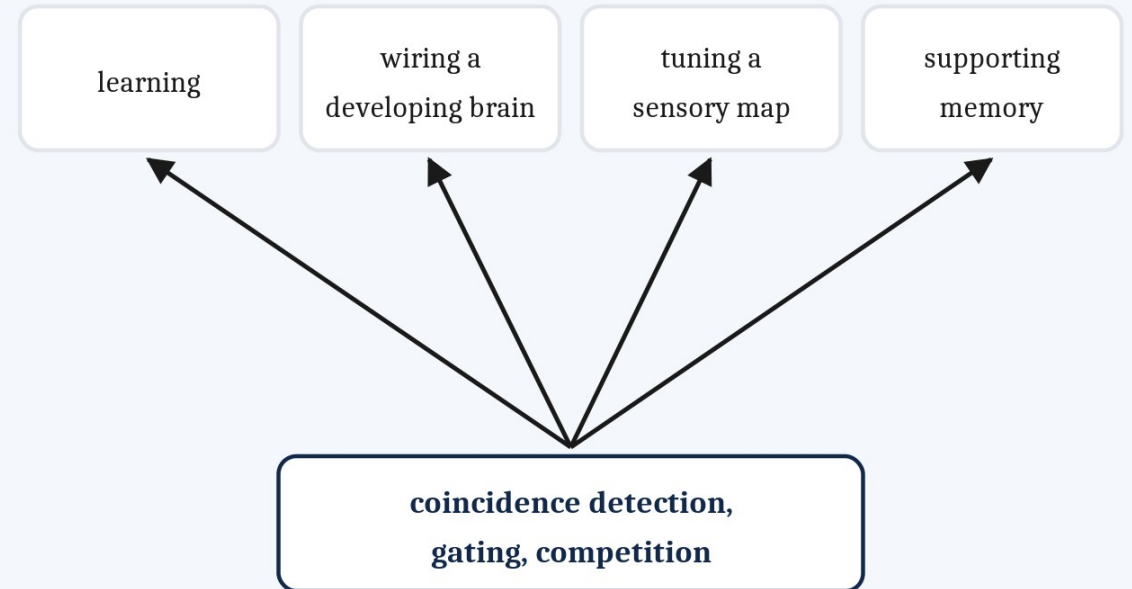
# One Mechanism, Several Jobs

The same coincidence detection, gating and competition also wire a developing brain, tune sensory maps and support memory. The converse of multiple realizability, and why **reverse inference** is unsafe.

## Multiple realizability (Module 1)



## The converse



Finding the mechanism does not tell us the job.

*Which is why reverse inference, from Module 2, is unsafe.*

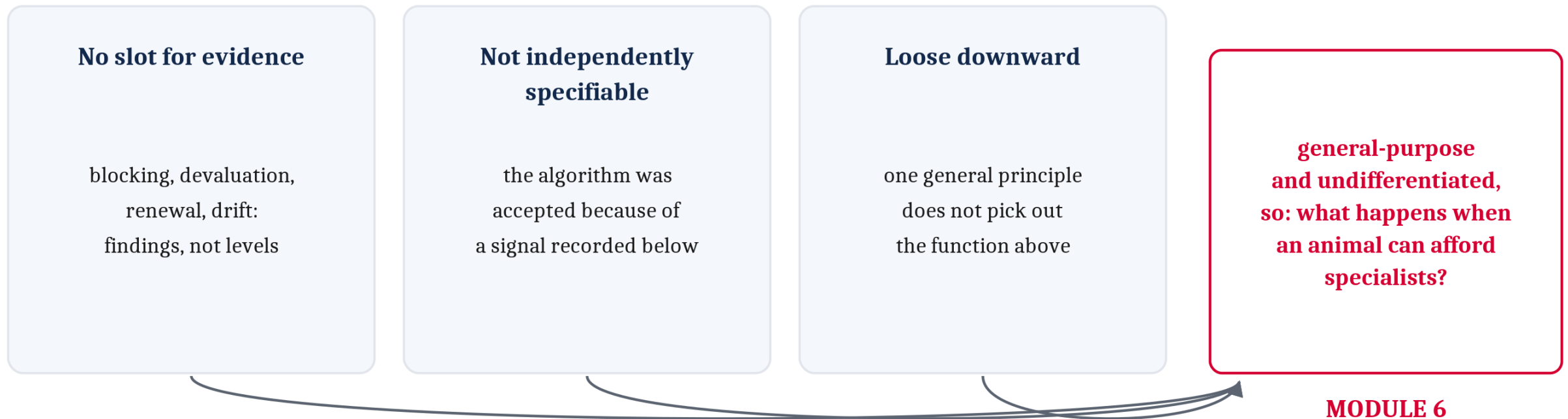
# What Separates the Four Kinds of Learning

Not what they change: all four change a weight. What separates them is how much of the world has to reach that weight before it moves, and who supplies the answer. One synapse, given the right third factor, implements all of it.

Vocabulary about goals and errors earns its place exactly where something in the mechanism plays the part the word names.

# Three Places Marr's Levels Strained, and What Comes Next

The levels have no slot for evidence. They were not independently specifiable here. And the implementation did not pick out one function. A principle reusable everywhere is specialized for nothing.



A framework is a tool with a shape; knowing where it strains is part of using it.

*Still waiting: Module 7 (a single unit's limits), Module 10 (records), Module 12 (models).*

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